AI4Life Calcium Imaging Denoising Challenge 2025

Denoising in Calcium Imaging

Fluorescence microscopy enables the visualization of dynamic biological processes at both cellular and subcellular scales, providing crucial insights into cellular structures, functions, and interactions of living cells.

Among the most widely used modalities is calcium imaging, which relies on genetically encoded calcium indicators (GECI) to indirectly measure intracellular calcium transients associated with cellular activity in neurons and other kinds of cells. These indicators fluoresce upon binding to calcium ions, allowing researchers to monitor functional activity across populations of living cells.

Calcium imaging data can be generally high-sensitive and bright, making it compatible with acquisition techniques such as two-photon microscopy. This allows measurements in highly scattering media, such as intact mammalian tissues, with good signal-to-noise ratios and without the need for averaging. Moreover, calcium imaging can capture the spiking activity of hundreds to thousands of neurons simultaneously, while maintaining single-cell resolution. This approach has been crucial in studying neuronal activity, synaptic transmission, and cellular signalling pathways.

Despite its advantages, calcium imaging presents several technical limitations. Its slower temporal resolution compared to other techniques such as voltage imaging makes it unsuitable for detecting fast events with high precision. As a result, while calcium imaging remains invaluable for certain

appliactions, it cannot provide a full picture of neuronal electrical activity. Moreover, the acquisition of high-quality, noise-free calcium imaging data is inherently difficult in practice due to noise in the acquired images. Noise in fluorescence microscopy arises from different sources, including photon shot noise and read noise, and can significantly mask the true biological signal.

Aim of the Challenge

The presence of noise in calcium imaging data poses a significant issue to signal reconstruction. This motivates the development of specialized denoising methods that are specific for this setting. In particular, calcium signals are both spatially and temporally structured.

Effective denoising methods must exploit information along both dimensions to preserve the dynamics of cellular activity,

with particular focus to preserve the temporal profile of the signal.

In this challenge, we highlight the importance of addressing both spatial and temporal reconstruction quality, and design evaluation metrics that explicitly quantify the denoising performances along these two dimensions.

Moreover, for a denoising method to be applicable in practice, it must generalize across heterogeneous image content (e.g, variation in morphology, fluorescence intensity, or background structure) and remain robust under different noise regimes. To address this, we designed a set of test scenarios with different noise levels and structural content.

An important limitation in denoising method research is the difficulty of obtaining clean ground truth data in real experiments. Even

when acquired in higher signal-to-noise conditions, the residual noise might make it difficult to use classical metrics for reconstruction quality estimation in a rigorous way. Recognising this, we designed this challenge to focus on synthetic datasets, were the ground truth signal is exactly known and noise can be introduced in a controlled way to ensure realistic conditions. This allows for a more precise and fair quantitative evaluation of algorithm performance.

Moreover, the same limitation applies not only to evaluation but also to model training. Supervised learning approaches rely on access to clean target data, which is typically scarcely available in real fluorescence ...

Calcium imaging provides data on cellular calcium

concentration, which reflects neuronal activity and other cellular processes like muscle contraction

Data provided by Calcium Imaging

- **Cellular Activity and Dynamics:**

It shows the fluctuations in intracellular calcium, an indirect marker for events such as action potentials in neurons and synaptic inputs.

- **Spatial Organization of Activity:**

The technique allows for precise spatial mapping of where and when cells are active, which is useful for understanding neural circuitry.

- **Population Activity:**

It can monitor the activity of

thousands of neurons simultaneously, providing a snapshot of overall network behavior.

- **Response Characteristics:**

Researchers can determine how cells respond to stimuli by measuring the amplitude and frequency of their calcium signals.

1. Simultaneous denoising, deconvolution, and demixing of

calcium imaging data

https://www.cell.com/neuron/fulltext/S0896-6273(15)01084-3

2. Reinforcing neuron extraction and spike inference in calcium imaging using deep self-supervised denoising

https://www.nature.com/articles/s41592-021-01225-0

3. Denoising two-photon calcium imaging data

https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0020490

4. Fast, efficient, and accurate neuro-imaging denoising via supervised deep learning

https://www.nature.com/articles/s41467-022-32886-w

5. Calcium imaging analysis–how far have we come?

https://pmc.ncbi.nlm.nih.gov/articles/PMC8406438/

6. Automated denoising software for calcium imaging signals using deep learning

https://www.cell.com/heliyon/fulltext/S2405-8440(24)15605-8

7. Analysis pipelines for calcium imaging data

https://www.sciencedirect.com/science/article/pii/S0959438818300941

8. **EZcalcium: open-source toolbox for analysis of calcium imaging data**

https://www.frontiersin.org/journals/neural-circuits/articles/10.3389/fncir.2020.00025/full

9. **Genetically encoded calcium indicators**

https://pubs.acs.org/doi/full/10.1021/cr078213v?casa_token=wEd9IT-7f5YAAAAA:-p8HsJagQMHCtiXX3OrRpinNmY7MpIHDXaz85CtWVgBmjBBdSs9PM7QE7SybkITAFOvSsgd51wr_3ovVBA